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Ahmad Asad

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ABOUT ME

Remote sensing scientist specializing in satellite-based algorithm development, cloud-native geospatial computing (GEE, Python), and artificial intelligence applications for environmental monitoring. Expertise includes multi-sensor integration (Sentinel-2, Landsat), spectral index formulation, sub-pixel analysis, and land-use/land-cover dynamics across contrasting agro-climatic landscapes.

EDUCATION AND TRAINING

21/10/2023 – CURRENT Italy

DOCTOR OF PHILOSOPHY IN SUSTAINABLE LAND MANAGEMENT University of Bari Aldo Moro

Satellite Remote Sensing and Radiative Transfer: Multi-sensor optical satellite integration (Sentinel-2, Landsat), pure-pixel extraction, spectral unmixing, spectral index formulation.
Machine Learning and GeoAI: Developing supervised and unsupervised machine learning classification pipelines for land-use and land-cover (LULC) change detection and multitemporal surface monitoring.
Cloud Geospatial Computing: Automated batch processing and big data satellite analysis using Google Earth Engine (GEE API scripting) and openEO.
Environmental and Land Degradation Analytics: Soil sealing quantification, urban heat island observation, and climatic impact assessment across contrasting agro-climatic landscapes.
Geospatial Engineering and Spatial Databases: Spatial database management (SQL/PostGIS), QGIS, Python data science libraries (GeoPandas, Rasterio, Scikit-learn), and interactive WebGIS prototyping.
Applied Energy and Infrastructure Mapping: Photovoltaic landscape modeling and spatial database administration using open territorial data.

Website https://www.uniba.it/it/dottorati/ | **Level in EQF** EQF level 8

15/02/2019 – 15/03/2022 Pakistan

MASTER OF REMOTE SENSING AND GIS COMSATS University

1. **Courses**: Advanced Remote Sensing, Image Processing, Geospatial Analysis, Customization, Geostatistical Analysis
2. **Workshops**: Time-Series Analysis using Google Earth Engine, Importance of RS/GIS in observing climate changes
3. **Conferences:** Environmental Degradation in Developing countries, Joint venture of France-Germany-Pakistan to find better solutions for Climate Change
4. **Projects:** Water quality analysis of Rawal Lake in Islamabad, Analysis of air quality around nurseries on Park Road, Islamabad, Detecting NDVI of Islamabad using Google Earth Engine, Snow cover analysis of Northern areas of Pakistan using MODIS imagery, Analysis of High Altitude areas of Pakistan using Landsat Imagery in Google Earth Engine.

Website https://www.comsats.edu.pk/ | **Field of study** Remote Sensing and Geographic Information Systems | **Final grade** 3.45 |

Level in EQF EQF level 7 | **Thesis** Analysis of Urban Heat Islands in High Altitude Areas of Pakistan

WORK EXPERIENCE

RESEARCH FELLOWSHIP – POLITECNICO DI BARI – 11/2026 – 05/02/2027 – BARI, ITALY

PHD SCHOLAR – UNIVERSITY OF BARI ALDO MORO – 21/10/2023 – Current

- The main theme of the PNRR funded PhD research is to enhance the accuracy of soil sealing detection using SRS tools.
- Development of a new Algorithm using Satellite imagery to detect sealed surfaces.

- Leveraging Machine Learning to enhance the accuracy of results.
- Observing climatic changes linked to Urbanisation, LULC monitoring, and time series analysis.
- Internee in the Administrative office to aid in geospatial analysis and geospatial database management.
- **International Mobility:** Research Scholar, University of Almería, Spain (Jan 2026 – Jul 2026) – Pure-pixel spectral analysis and spectral unmixing workflows.

UNIVERSITY RESEARCH ASSISTANT – COMSATS UNIVERSITY – 02/02/2022 – 10/10/2023 – PAKISTAN

TECHNICAL WRITER – CLOUD INNOVATOR SOLUTIONS – 01/01/2019 – 01/09/2023 – PAKISTAN

LANGUAGE SKILLS

Mother tongue(s): **URDU**

Other language(s):

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLISH	C1	C2	C1	C1	C2
ITALIAN	A2	A2	A2	A1	A2

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

SKILLS

Machine Learning | Google Earth Engine | Arc-GIS | Javascipt | Python | QGIS | ENVI | Geospatial data entry | GeoDa (user level) | R | principles of artificial intelligence

PUBLICATIONS

2026

A multitemporal Sentinel-2 framework leveraging machine learning for urban sealing detection across climatic regions

Authors: Ahmad Asad, Alessandra Capolupo & Eufemia Tarantino | **Journal Name**: Remote Sensing Applications: Society and Environment | **Volume, Issue and Pages**: Volume 43, 102231 | **Publisher**: Elsevier

2027

Mapping the Photovoltaic Landscape of Puglia Using Open Energy Data

Authors: Ahmad Asad, Alessandra Capolupo & Eufemia Tarantino | **Journal Name**: Computational Science and Its Applications – ICCSA 2026 Workshops | **Volume, Issue and Pages**: LNCS,volume 16762 | **Publisher**: Springer

2026

SoilEye: An Interactive Web App Prototype for Soil Sealing Monitoring

Authors: Ahmad Asad, Alessandra Capolupo & Eufemia Tarantino | **Journal Name**: Computational Science and Its Applications – ICCSA 2026 Workshops | **Volume, Issue and Pages**: LNCS,volume 15891 | **Publisher**: Springer

HONOURS AND AWARDS

PNRR Doctoral Research Fellowship – Italian Ministry of University and Research (MUR) / NextGenerationEU (2023 – Present)

PRIN Research Fellowship (EO4DES) – Politecnico di Bari